Universidade Leiria • Oeste
Escola de Tecnologia e Gestão

TrackSeed: Augmented Reality Cone Placement and a Track Digital Twin for Formula Student – State of the Art and Project Proposal

André Lopes, Tomás Santos
School of Technology and Management, University of Leiria and Oeste, Leiria, Portugal
2023123014@ulo.pt, 2023122551@ulo.pt
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***Abstract*—[TODO: 150–200 words, written last. Cover: (1) the problem – Formula Student tracks with hundreds of cones are set out by hand with tapes and trigonometry, which is slow and error-prone, and driverless cars depend on the layout matching the plan; (2) the review scope – mobile AR for on-site layout, AR tracking accuracy and alignment, shared multi-user AR, digital twins and real-time platforms; (3) how the literature was selected; (4) the main findings, e.g. which alignment approaches reach the required accuracy and the gaps; (5) the resulting proposal, TrackSeed. State expected benefits as hypotheses; report no results.]**

***Index Terms*—augmented reality; digital twin; mobile AR tracking; spatial layout; Formula Student; real-time synchronisation**

I. Introduction

Formula Student (FS) is an international engineering competition in which university teams design, build and race a single-seat car; in the driverless (DV) class, the same car must complete the dynamic events autonomously. Every dynamic event is driven on a track that exists only as a pattern of traffic cones on an open area of tarmac. Under the 2026 driverless specification, the left border is marked with small blue cones, the right border with small yellow cones, entry and exit lanes with small orange cones, and big orange cones are placed before and after the start, finish and timekeeping lines, with at most 5 m between consecutive cones in the driving direction and shorter spacing in corners [1]. Teams reproduce these layouts many times per season on their own test sites: acceleration and skidpad layouts for vehicle tuning, and autocross and trackdrive circuits of a few hundred metres, which already need well over a hundred cones per lap.

These tracks are usually set out by hand. A plan drawn on paper or in a spreadsheet is transferred to the ground with measuring tapes and trigonometry, which needs several people, takes time away from testing and provides no check of the result: a misplaced or wrong-coloured cone is only corrected if someone notices it. For driverless cars, the consequences go beyond inconvenience. Their perception and mapping pipelines use the cones as the landmarks that define the track, and path planning and control are built on the map estimated from them [2]. A layout that differs from the intended one therefore changes the problem the car is solving and makes results from different test days hard to compare. Surveying instruments such as total stations and real-time kinematic (RTK) global navigation satellite system (GNSS) receivers can set out points with centimetre accuracy, but they are expensive and require trained operators [TODO: cite a surveying source]. None of these options links the plan, the placement on the field and a record of what was actually built.

Augmented reality (AR) combines virtual content with the user's view of the real world, registered in three dimensions and interactive in real time [3]. Mobile AR frameworks on consumer smartphones, such as Google's ARCore, estimate the pose of the device by combining visual feature points from the camera with inertial measurements [4]. This makes it possible to show each planned cone, with its colour, at its position on the ground through the phone that team members already carry. Compared with the conventional alternatives (a tape measure, or a 2D map showing the phone's GNSS position), AR guidance removes the measuring step, shows the instruction where the cone must be placed, and allows the same camera to verify the placement. Evidence from construction point layout shows both the potential and the main limitation: with a head-mounted AR device registered to a paper marker, Chalhoub and Ayer reported a locational accuracy of about 5 cm, which became worse further from the marker [5]. Their indoor test area was much smaller than an FS test site, so this accuracy cannot be assumed to hold over a full track.

The proposed system, TrackSeed, combines the contributions of the two course units around a shared model of the track. Virtual and Augmented Reality (VAR) provides the AR guidance aligned to the real ground and a digital twin (DT) of the track. Following Kritzinger et al. [6], the term DT is used here only for a virtual counterpart that exchanges data automatically with its physical counterpart in both directions:

the TrackSeed twin receives every confirmed cone from the phones and sends track edits and section assignments back to them. Emerging and Intelligent Technologies (EIT) provides the cloud platform that stores track plans and placement sessions, real-time synchronisation between the phones and a web dashboard, remote configuration of the devices, and on-device intelligence that checks each placed cone with the camera.

This paper reviews the state of the art that TrackSeed builds on, with three objectives: (i) to establish what accuracy mobile AR achieves for on-site layout and guidance, and which alignment approaches limit drift over large outdoor areas; (ii) to compare architectures for a bidirectional DT synchronised in real time between mobile and web clients; and (iii) to derive and justify the technology choices and validation plan of the proposal. Section II reviews the literature by theme, Section III analyses the candidate technologies, Section IV presents the proposed solution and its validation plan, Section V discusses feasibility and limitations, and Section VI concludes.

A. Scope and literature search

The literature was searched in IEEE Xplore, Scopus, the ACM Digital Library and Google Scholar [TODO: confirm the databases actually used and the search dates]. Searches combined "augmented reality" with "construction layout", "stakeout" or "setout"; "ARCore" or "mobile AR" with "accuracy" or "drift"; "collaborative AR" with "shared anchors"; and "digital twin" with "classification" or "real-time synchronisation". Peer-reviewed journal and conference papers published from 2015 onwards that report a quantitative evaluation with handheld or head-mounted devices were preferred. Foundational works were kept for definitions, and official documentation and the FS rules were used only for tool capabilities and domain requirements, not as evidence of performance. Since almost no studies address track setout in motorsport, the review relies mainly on adjacent domains, construction layout in particular, whose tasks, distances and accuracy requirements differ from those of an FS test site. This limits how directly their results transfer, as discussed in Section V. [TODO: number of records screened and selected]

II. State of the Art

A. AR for on-site layout and spatial guidance

[TODO: Closest neighbour domain: AR-guided construction layout/setout, surveying stakeout, AR navigation to a target point. Compare what accuracy studies achieved, over what distances, on which devices, and with what ground truth. Distinguish measured accuracy from authors' expectations. 3–5 studies.]

B. Mobile AR tracking accuracy and alignment to the ground

[TODO: How ARCore/ARKit track (visual-inertial odometry) and what studies report on drift over distance and on low-texture surfaces. Compare alignment approaches: fiducial markers (ArUco, image targets), geospatial/VPS localisation, GNSS and RTK GNSS fusion. Note why heading error matters more than position error over long tracks (1 degree is about 3.5 m at 200 m). Use studies for accuracy claims and official docs only for capabilities.]

C. Shared and multi-user AR

[TODO: How several devices share one coordinate frame (cloud/shared anchors, marker-based co-localisation); reported alignment error and latency between devices; implications for several people placing cones at once.]

D. Digital twins

[TODO: Origin and definitions (Grieves; Kritzinger et al. 2018 classification of digital model, digital shadow and digital twin by direction of data flow – verify citation). Argue where TrackSeed sits: phones update the twin (physical to digital) and remote edits by the team lead are pushed back to the phones (digital to physical), so it is a twin rather than a shadow. Compare DT examples in sport, construction or site monitoring.]

E. Real-time platforms and synchronisation

[TODO: Architectural approaches for live sync between mobile clients and a web dashboard: WebSocket, MQTT publish/subscribe, managed real-time databases; offline-first sync and conflict handling; cloud vs on-device processing. Reported latency and scalability evidence.]

F. Cone perception in Formula Student

[TODO: Brief context for the intelligence component: FSOCO dataset (Vödisch et al. – verify), cone detection in FS driverless pipelines (e.g. AMZ real-time 3D cone detection – verify). Note the domain shift between car-mounted and handheld phone cameras. Keep this short: the ML depth belongs to the MDL course.]

G. Comparative synthesis

[TODO: Explain the patterns, disagreements and gaps in Table I: e.g. accuracy is mostly reported over short distances; few studies evaluate large outdoor areas; multi-user setups are rarely validated on site; no work combines placement guidance, verification and a live twin. End with the gap TrackSeed addresses.]

III. Technological Analysis

A. Aligning the track plan to the ground

[TODO: Compare the options in Table II against the requirement (provisional target: about 10–15 cm per cone over tracks of 200 m or more). Justify the choice, e.g. two-point marker alignment with periodic re-alignment, with RTK GNSS as an optional higher-accuracy reference; explain why the geospatial API alone is insufficient. Open decision: how the reference markers themselves are surveyed.]

TABLE I
COMPARISON OF THE REVIEWED LITERATURE

Study	Theme	Context and approach	Measured evidence	Limitations	Decision informed
[TODO: ref]	AR layout / setout	[TODO:]	[TODO: accuracy, distance, device]	[TODO:]	Guidance design, target accuracy
[TODO: ref]	AR layout / setout	[TODO:]	[TODO:]	[TODO:]	[TODO:]
[TODO: ref]	AR tracking drift	[TODO:]	[TODO: drift vs distance]	[TODO:]	Re-alignment strategy
[TODO: ref]	Ground alignment	[TODO:]	[TODO:]	[TODO:]	Alignment method
[TODO: ref]	Shared AR	[TODO:]	[TODO: inter-device error, latency]	[TODO:]	Multi-user design
[TODO: ref]	Digital twin	[TODO:]	[TODO:]	[TODO:]	Twin data model and data flows
[TODO: ref]	Real-time sync	[TODO:]	[TODO: latency, scale]	[TODO:]	Communication protocol
[TODO: ref]	FS cone perception	[TODO:]	[TODO:]	[TODO:]	On-device verification

TABLE II
APPROACHES FOR ALIGNING THE TRACK PLAN TO THE REAL GROUND

Approach	Reported accuracy	Range and drift	Cost	Setup effort	Multi-user	Sources
Tape and trigonometry (baseline)	[TODO:]	No drift; error accumulates per measurement	Very low	High (several people)	Yes (manual)	[TODO: ref]
Single fiducial marker (ArUco / image target)	[TODO:]	[TODO: heading error grows with distance]	Low	Low	Per device	[TODO: ref]
Two-point marker alignment + re-alignment	[TODO:]	[TODO:]	Low	Medium (survey markers)	Per device	[TODO: ref]
Geospatial / VPS localisation	[TODO: metre level?]	[TODO:]	Free (API)	Low	Yes	[TODO: ref]
RTK GNSS reference	[TODO: cm level]	No visual drift; needs sky view	Medium (receiver)	Medium	Yes	[TODO: ref]
Shared cloud anchors	[TODO:]	[TODO:]	Free (API)	Low	Yes	[TODO: ref]

B. Mobile AR platform

[TODO: Justify Android native (Kotlin + ARCore, SceneView for rendering) against Unity AR Foundation and ARKit: device availability within the team, access to camera frames for on-device inference, native sensors and offline storage, development effort. Note the main compromise (more rendering effort, Android only) and how the web dashboard covers AnyAccess.]

C. Platform and real-time communication

[TODO: Compare the options in Table III and justify the choice for phone–cloud–dashboard synchronisation, offline mode with later sync, and remote configuration pushed to devices. Cover where computation happens (on-device inference vs cloud), containerised deployment (Docker) and the open JSON track schema for interoperability.]

D. Digital twin and dashboard

[TODO: Justify the twin representation (2D map vs 3D scene, e.g. React with Three.js), which state it holds per cone (planned, placed, verified, wrong colour, missing) and how it is updated. Open decisions still to make.]

TABLE III
REAL-TIME COMMUNICATION OPTIONS

Option	Latency	Offline support	Scalability	Effort
REST polling	[TODO:]	[TODO:]	[TODO:]	Low
WebSocket	[TODO:]	[TODO:]	[TODO:]	[TODO:]
MQTT	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Managed real-time DB	[TODO:]	[TODO:]	[TODO:]	[TODO:]

IV. PROJECT PROPOSAL

A. Solution and conceptual architecture

[TODO: Users and usage scenario: the team lead uploads a track JSON; placers on the field follow AR markers to each cone position; the phone verifies each placed cone and its colour; the dashboard twin fills in live, highlights wrong or missing cones and allows remote edits and section assignment.]

[TODO: Objectives, essential functionalities and boundaries (what is out of scope this semester, e.g. iOS, the geospatial API as primary alignment).]

[TODO: Explain Fig. 1: components (track import and validation, cloud API and twin store, real-time sync service, Android app with AR guidance and on-device verification, web dashboard), labelled data flows, where computation happens, and the contribution of each course unit (EIT, VAR, DMD, MDL). List equipment, software, data and service dependencies. Mark provisional choices.]

[TODO: Preliminary plan: October – track format and AR alignment on a small track; November – on-device verification, cloud platform and sync; December – dashboard twin, multi-user, full track tests; January – validation and report.]

B. Preliminary validation

[TODO: Introduce Table IV. For each metric, describe the test, resources or participants, baseline and provisional success criterion. Label expected benefits as hypotheses and distinguish this plan from the evaluations reported by the reviewed studies.]

TABLE IV
PLANNED EVALUATION CRITERIA

Metric	**Test**	**Baseline**	**Provisional target**
Placement error (cm)	Place a skidpad/autocross section; measure each cone against tape or RTK ground truth	Tape-measure setout	[TODO: e.g. $\leq$15 cm median]
Setup time (min)	Same track set out by the same team with each method	Tape-measure setout	[TODO: e.g. 30% faster]
Sync latency (ms)	Time from cone confirmation on the phone to its appearance on the dashboard	–	[TODO: e.g. $\leq$1 s]

V. Discussion

[TODO: Feasibility given two students, one semester and available phones/fields. Main risks: AR drift and heading error over long tracks, low-texture asphalt, surveying the reference markers, multi-device alignment, outdoor lighting for verification, winter weather for field tests. Limitations of the evidence (short distances, controlled conditions, adjacent domains). Concerns such as user safety on track and outdoor screen readability.]

[TODO: How the review changed the proposal (e.g. two-point alignment instead of a single marker; geospatial API demoted to fallback), rejected alternatives (Unity, iOS, total station) and when decisions may need revision. State the expected contribution as a justified integration and evaluation of existing technologies, not unproven novelty.]

VI. Conclusions

[TODO: Summarise the review findings, the resulting decisions and the expected contribution; return to the objectives; name the next development and validation steps. No new evidence and no claimed results.]

AI Use Declaration

[TODO: Write a truthful declaration. If kept: "The authors used Claude (Anthropic, Claude Opus 5.5) for [e.g. analysing the assignment brief, proposing the paper outline and table structure, and reviewing drafts] in [sections]. All outputs were reviewed, and references and technical claims were checked against their sources. The authors remain responsible for the submitted work." Update the list as the work progresses.]

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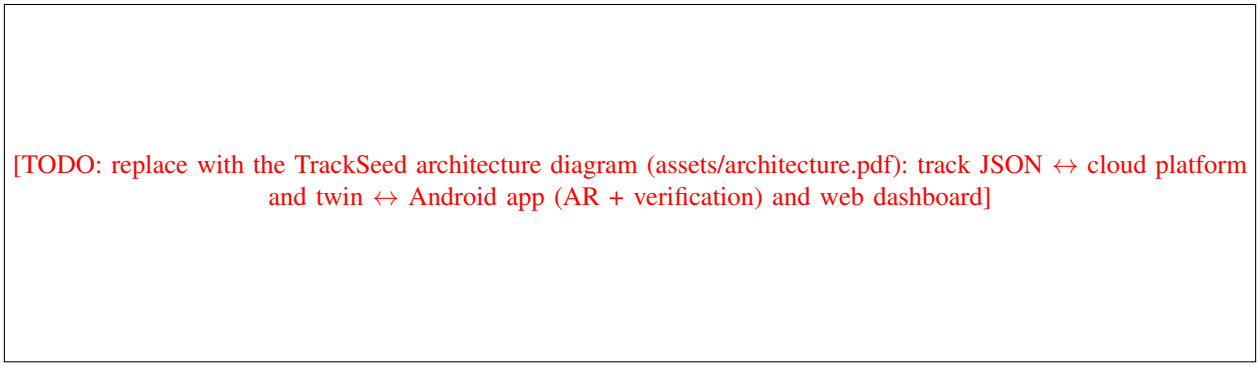


Fig. 1. Conceptual architecture of TrackSeed. [TODO: explain the arrows and data exchanged.]